

Qatar-1b

R-2778

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_140809 · created 2026-09-03T23:49:33+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456878.79437 +/- 0.00035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.012 +/- 0.053
Transit depth (Rp/Rs)^2	0.014 +/- 0.13 [%]
Orbital Inclination (inc)	79.27 +/- 2.86
Airmass coefficient 1 (a1)	3.66 +/- 1.77
Airmass coefficient 2 (a2)	-1.02 +/- 1.3
Scatter in the residuals of the lightcurve fit is	49.37 %
Best Comparison Star	#2 - [628, 75]
Optimal Aperture	1.71
Optimal Annulus	6.91
Transit Duration (day)	0.013 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.012 ± 0.053 vs 0.1463 (deviation 2.53σ)
Duration fit vs published	0.013 d vs 0.0692 d (deviation 2.67σ)
Mid-time O–C	0.2 min
Depth SNR	0.2
Residual scatter	49.37 %

Light curve

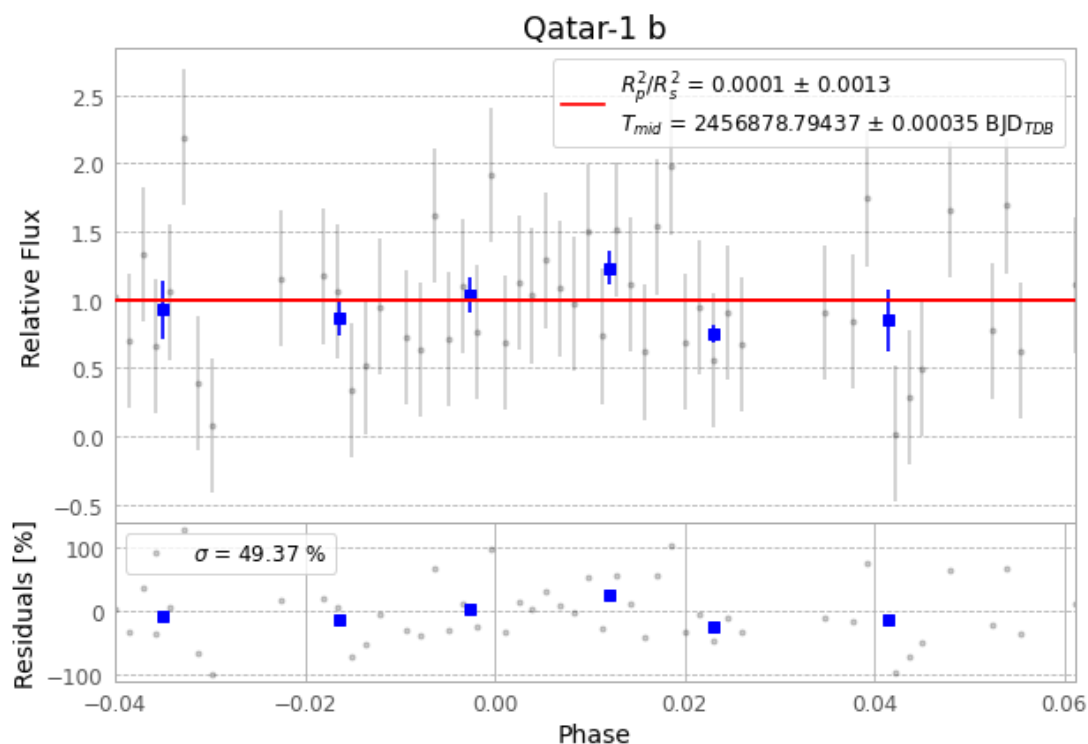


Figure 1 — FinalLightCurve_Qatar-1 b_2014-08-08.png (SHA-256 8042ed563e2bce37...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [368, 327], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cf760b0437c7ba1f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

69 FITS frames (45 MB), Qatar-1140809053913.FITS → Qatar-1140809090612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-140809.log (37b124ac379e...), FinalLightCurve_Qatar-1 b_2014-08-08.png (8042ed563e2b...), FinalLightCurve_Qatar-1 b_2014-08-08.csv (76f7e8612959...)

Compute resources

Wall time 373.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 23:49Z.